

NIZHNEUDINSK, Russian Federation

WMO: 296980

Lat: 54.883N

Lon: 99.033E

Elev: 411

StdP: 96.48

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.2	(c) -35.1	(d) -40.4	(e) 0.1	(f) -38.2	(g) -37.3	(h) 0.1	(i) -35.0	(j) 7.3	(k) -11.0	(l) 6.1	(m) -11.6	(n) 0.2	(o) 100	(p) 0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.4	(c) 29.0	(d) 18.7	(e) 27.2	(f) 18.0	(g) 25.5	(h) 17.1	(i) 20.1	(j) 26.5	(k) 19.1	(l) 25.1	(m) 18.2	(n) 23.7	(o) 2.2	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.6	22.4	17.0	12.8	21.7	16.1	12.0	20.8	59.7	26.6	56.4	25.1	53.5	23.9	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.5	(b) 5.7	(c) 4.5	(e) -40.3	(f) 32.7	(g) 4.3	(h) 1.7	(i) -43.4	(j) 33.9	(k) -45.9	(l) 34.9	(m) -48.3	(n) 35.9	(o) -51.4	(p) 37.2		
(a) 7.5	(b) 5.7	(c) 4.5	(e) -40.2	(f) 22.0	(g) 4.2	(h) 1.5	(i) -43.3	(j) 23.1	(k) -45.7	(l) 24.0	(m) -48.1	(n) 24.8	(o) -51.1	(p) 26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.2	-20.4	-15.9	-6.6	2.5	9.0	15.7	18.2	15.7	8.7	1.1	-9.3	-17.6		
	DBStd	14.41	7.60	7.40	6.72	4.85	4.68	3.77	2.66	3.34	4.20	5.02	7.44	7.89		
	HDD10.0	4276	943	724	516	228	77	4	0	2	73	276	578	854		
	HDD18.3	6683	1202	957	774	475	288	95	36	93	289	533	828	1113		
	CDD10.0	693	0	0	0	3	47	174	253	180	35	1	0	0		
	CDD18.3	59	0	0	0	0	1	15	31	12	1	0	0	0		
	CDH23.3	923	0	0	0	2	67	308	341	186	20	0	0	0		
	CDH26.7	211	0	0	0	1	11	79	77	40	3	0	0	0		
	WSAvg	1.5	1.0	1.2	1.7	2.2	2.2	1.6	1.3	1.3	1.5	1.7	1.5	1.1		
	PrecAvg	404	10	9	7	17	34	46	107	83	48	21	12	10		
(15) Precipitation	PrecMax	540	33	22	27	47	64	100	180	160	85	52	28	18		
	PrecMin	258	2	1	1	2	7	14	18	28	15	1	2	5		
	PrecStd	75	7	6	6	12	14	24	40	29	21	15	7	4		
	PrecStd	75	7	6	6	12	14	24	40	29	21	15	7	4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.3	3.5	11.2	22.0	28.2	31.5	31.6	30.2	26.0	17.9	8.3	1.2		
		MCWB	-2.3	-0.5	4.6	10.7	15.6	18.8	21.2	19.6	16.2	9.2	3.8	-1.7		
	2%	DB	-3.3	0.3	8.0	17.7	25.4	28.9	28.7	27.5	21.9	14.3	4.6	-1.8		
		MCWB	-4.4	-2.3	2.7	8.4	14.0	18.0	19.4	18.9	14.1	8.2	1.6	-3.5		
	5%	DB	-6.3	-2.4	5.4	14.6	22.5	26.9	26.9	25.4	18.9	11.0	2.4	-4.2		
		MCWB	-7.5	-4.3	1.1	7.0	12.4	17.2	18.7	18.0	12.7	6.3	0.2	-5.5		
	10%	DB	-9.3	-4.9	3.2	11.8	19.6	24.7	25.1	23.1	16.0	8.2	0.4	-6.5		
		MCWB	-10.2	-6.5	-0.2	5.6	11.1	16.4	17.9	16.8	11.3	4.7	-1.2	-7.5		
	0.4%	WB	-2.0	-0.1	5.1	11.0	16.8	20.9	22.1	21.2	17.1	10.5	4.3	-1.4		
		MCDB	-0.9	2.9	10.5	20.5	25.8	28.6	28.3	27.5	23.7	16.1	7.5	0.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-4.6	-2.3	3.1	9.2	14.6	19.2	20.3	19.5	14.8	8.6	1.8	-3.4		
		MCDB	-3.3	0.1	7.5	16.0	23.8	26.3	26.6	25.9	20.5	13.3	4.1	-1.8		
	5%	WB	-7.5	-4.3	1.4	7.4	13.1	18.1	19.5	18.4	13.3	6.7	0.3	-5.5		
		MCDB	-6.3	-2.3	4.9	14.2	21.1	24.8	25.1	24.3	17.7	10.7	2.0	-4.3		
	10%	WB	-10.2	-6.4	0.0	5.7	11.4	17.0	18.6	17.3	11.9	5.0	-1.1	-7.5		
		MCDB	-9.3	-4.9	2.9	11.4	18.6	23.2	23.6	22.1	15.2	7.9	0.4	-6.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	14.8	14.6	12.5	13.6	13.4	11.4	11.4	11.2	10.3	9.0	9.3		
		MCDBR	11.3	15.2	16.7	18.9	20.2	18.3	15.5	16.3	17.0	15.7	10.5	10.4		
		MCWBR	9.9	12.6	11.7	10.7	10.0	8.2	7.0	7.8	9.0	9.4	7.7	8.9		
		MCDBR	11.1	14.7	14.8	17.3	18.1	15.4	12.9	14.4	15.1	14.8	9.8	9.9		
	5% WB	MCWBR	9.8	12.1	10.4	10.2	9.3	7.5	6.4	7.3	8.2	9.1	7.4	8.7		
		MCWBR	9.8	12.1	10.4	10.2	9.3	7.5	6.4	7.3	8.2	9.1	7.4	8.7		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.313	0.350	0.388	0.419	0.419	0.426	0.403	0.354	0.336	0.308	0.282		
	taud		2.059	2.027	2.045	2.178	2.159	2.260	2.273	2.316	2.405	2.314	2.147	2.017		
	Ebn at Noon		625	736	801	830	824	828	814	811	808	722	584	513		
	Edh at Noon		68	101	126	128	138	127	123	111	89	76	62	57		
(44) All-Sky Solar Radiation	RadAvg		0.95	1.95	3.38	4.52	5.12	5.79	5.25	4.42	3.07	1.75	0.92	0.62		
	RadStd		0.07	0.10	0.18	0.25	0.46	0.37	0.44	0.30	0.30	0.14	0.10	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a) Station Only	(b) Regional (6 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(a) Station Only	(b) Regional (6 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A

Nomenclature: See separate page